

What is FAIR and AI-Ready for Datasets?

Bhavesh Patel, Ph.D.
Research Professor



CALIFORNIA MEDICAL
INNOVATIONS INSTITUTE

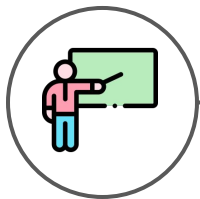


FAIR DATA
INNOVATIONS HUB

About This Presentation



20 min



How to make research
data FAIR and AI-ready

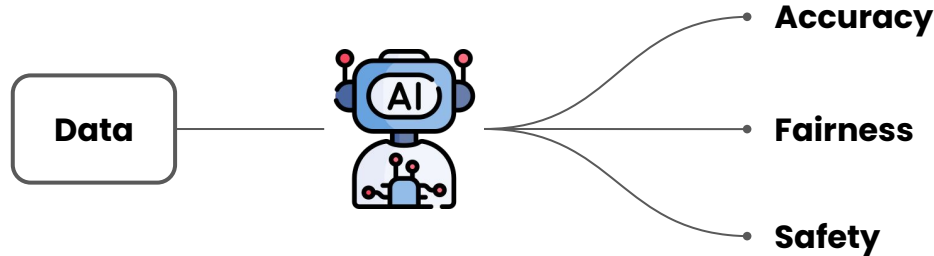


AI-Ready Data

AI-Ready Data

Data is the fuel for AI

Data **enables the development** of AI models
and directly **affects their characteristics**

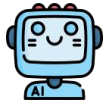


AI-Ready Data

Making research data AI-ready is important

If you share your research data
it is very **likely to reused for AI development**

By making your research data AI ready you can:



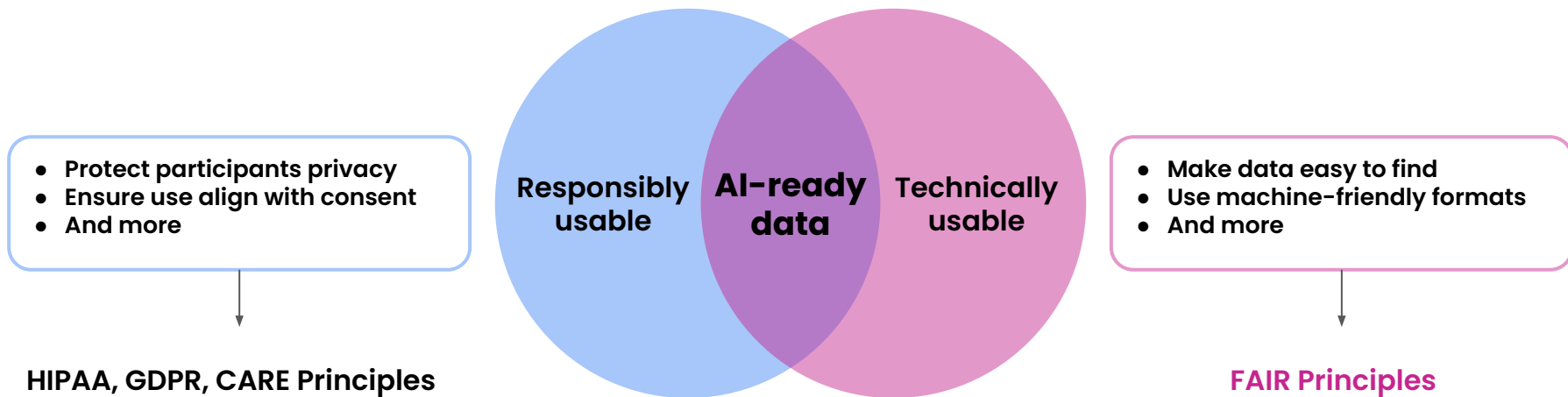
Support and facilitate the development of AI models



Ensure that your data is not (un)intentionally misused

AI-Ready Data

How to make data AI-ready?





FAIR Principles

FAIR Principles

Origin

How to make all research outcomes, including data,
optimally reusable by humans and machines?



**Findable, Accessible, Interoperable, and Reusable (FAIR)
Principles (2016)**

FAIR Principles

15 principles to optimize data reuse for humans and machines

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:

- A1. (meta)data are retrievable by their identifier using a standardized communications protocol
 - A1.1 the protocol is open, free, and universally implementable
 - A1.2 the protocol allows for an authentication and authorization procedure, where necessary
- A2. metadata are accessible, even when the data are no longer available

To be Interoperable:

- I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (meta)data use vocabularies that follow FAIR principles
- I3. (meta)data include qualified references to other (meta)data

To be Reusable:

- R1. meta(data) are richly described with a plurality of accurate and relevant attributes
 - R1.1. (meta)data are released with a clear and accessible data usage license
 - R1.2. (meta)data are associated with detailed provenance
 - R1.3. (meta)data meet domain-relevant community standards

FAIR Principles

Interpretation



Findable → Data are easy to find

Data have a unique identifier (e.g. a DOI)
Metadata are provided and indexed in search engines



Accessible → Data access process is clear

Data are accessible using a standard protocol (e.g., HTTP)
Access requirements are clearly stated



Interoperable → Data work well with related data/tools

Data and metadata are in standard, machine-friendly format
Data are connected to related resources



Reusable → Data is easily usable

Data have a clear usage license
Data are well documented

FAIR Principles

What they are not

FAIR  Open

FAIR  Ethical or fair

FAIR Principles

Adoption

The G20 logo consists of the letters "G20" in a bold, black, sans-serif font, centered within a light blue circle. This circle is itself centered within a larger, light gray rounded square.

G20

Leaders at the 2016 G20 meeting released a joint press release expressing their intention to support implementation of FAIR principles in publicly funded research



“Turning FAIR into Reality”
report (2018)

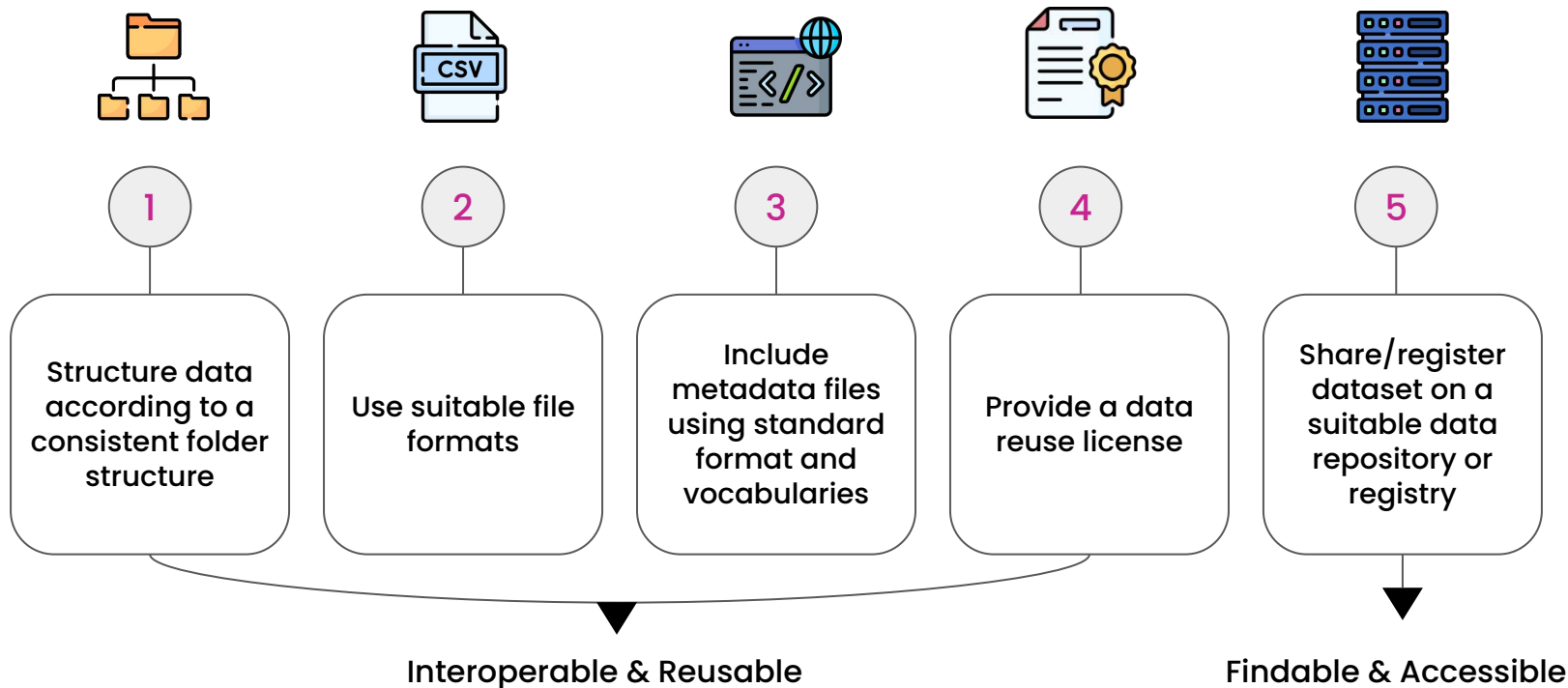


National Institutes
of Health

New data sharing policy (January 2023) requires all proposal to include a Data Management and Sharing Plan (DMSP) that describes how data will be made FAIR

FAIR Principles

How to practically make data FAIR



The AI-READI Dataset



The AI-READI Dataset

About AI-READI

AI-READI is one of the **Data Generation Project (DGP)**
of the NIH Bridge2AI Common Fund program

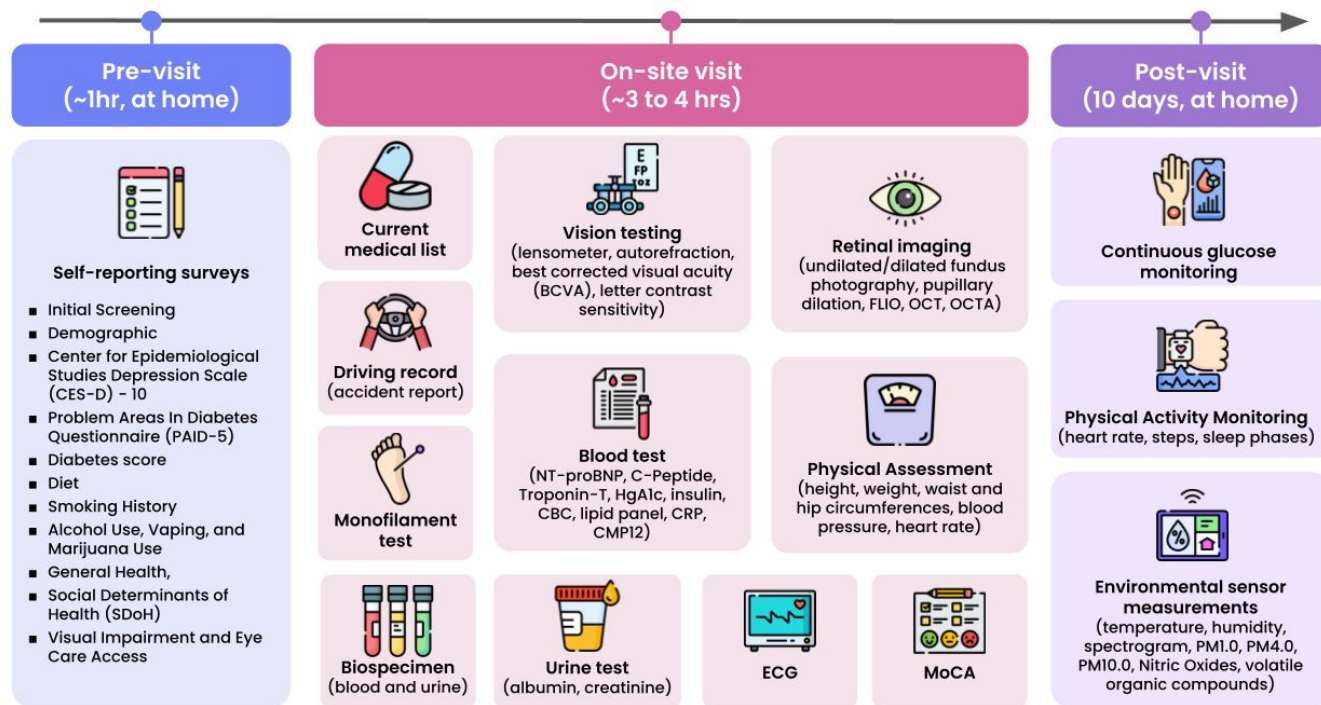


Goal: collect a multimodal dataset for studying
Type 2 Diabetes and make it AI-ready

<https://aireadi.org>

The AI-READI Dataset

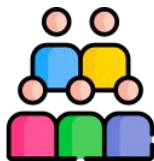
Data collection protocol



FLO = Fluorescence Lifetime Imaging, OCT = Optical Coherence Tomography, OCTA = Optical Coherence Tomography Angiography,
ECG = Electrocardiogram, MoCA = Montreal Cognitive Assessment, PM1.0, 4.0, and 10.0 = Particulate matter less than 1, 4, and 10 microns, respectively

The AI-READI Dataset

Version 2 (November 2024)



Data from 1067
participants



165,000+ files



2 TB of data total

Target by the end of the project: 4,000 participants

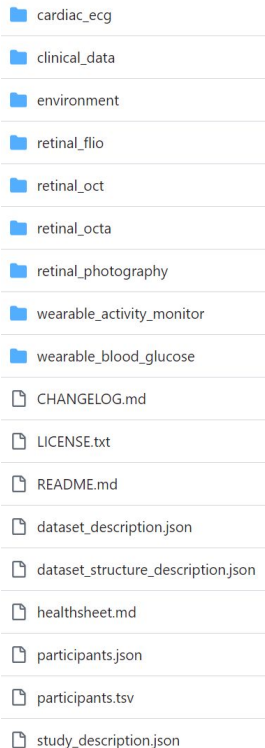
<https://fairhub.io/datasets/2>

The AI-READI Dataset

FAIR: 1. Consistent folder structure

Clinical Dataset Structure (CDS)

cds-specification.readthedocs.io



One folder per
datatype

Metadata files

The AI-READI Dataset

FAIR: 2. Standard data format

Data category	Data format followed
Clinical data	Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)
Retinal imaging data	Digital Imaging and Communications in Medicine (DICOM)
ECG data (standard 12-lead)	WaveForm DataBase (WFDB)
Wearables data (physical activity monitoring, continuous glucose monitoring data)	Open mHealth
Environmental sensor data	Earth Science Data Systems (ESDS) format

The AI-READI Dataset

FAIR: 3. Extensive metadata

retinal_photography

wearable_activity_monitor

- CHANGELOG.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- LICENSE.txt
- participants.json
- participants.tsv
- README.md
- study_description.json

Broad metadata: study background, dataset provenance, dataset structure, license terms, participants information

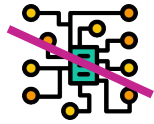
Provided in both human and machine-friendly formats

The AI-READI Dataset

FAIR: 4. Shared under a new usage license



Prohibits re-identification of study participants or harm



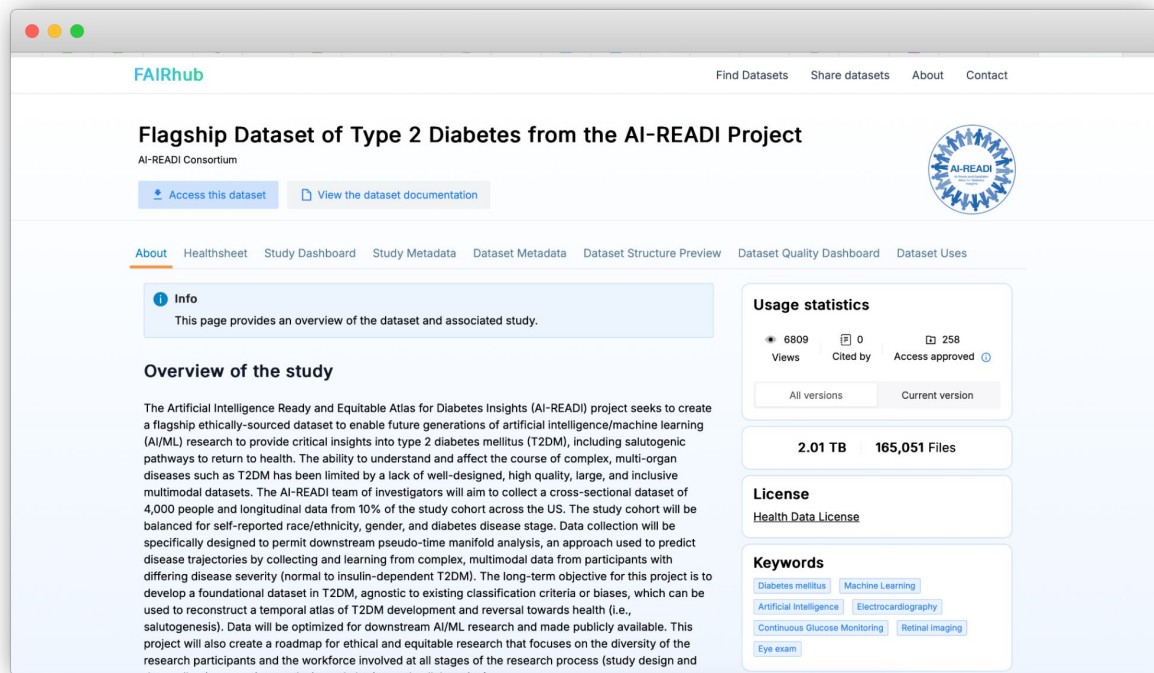
Prohibits distribution of the data and of models that may “copy” or “memorize” the data



Allows commercial use

The AI-READI Dataset

FAIR: 5. Shared on FAIRhub



The screenshot shows the FAIRhub website interface for the 'Flagship Dataset of Type 2 Diabetes from the AI-READI Project'. The page includes a navigation bar with links to 'Find Datasets', 'Share datasets', 'About', and 'Contact'. Below the header, there's a section for the dataset title and a link to 'Access this dataset'. A horizontal menu lists various tabs: 'About', 'Healthsheet', 'Study Dashboard', 'Study Metadata', 'Dataset Metadata', 'Dataset Structure Preview', 'Dataset Quality Dashboard', and 'Dataset Uses'. The 'About' tab is active, showing an 'Info' section with a brief overview of the dataset and associated study. Below this is the 'Overview of the study' section, which provides a detailed description of the AI-READI project, its goals, and the dataset's characteristics. On the right side, there's a 'Usage statistics' section showing 6809 views, 0 citations, and 258 access approvals. Below this, it displays '2.01 TB' and '165,051 Files'. The 'License' section indicates 'Health Data License'. The 'Keywords' section lists terms like 'Diabetes mellitus', 'Machine Learning', 'Artificial Intelligence', 'Electrocardiography', 'Continuous Glucose Monitoring', 'Retinal imaging', and 'Eye exam'.

FAIRhub

Find Datasets Share datasets About Contact

Flagship Dataset of Type 2 Diabetes from the AI-READI Project

AI-READI Consortium

[Access this dataset](#) [View the dataset documentation](#)

[About](#) [Healthsheet](#) [Study Dashboard](#) [Study Metadata](#) [Dataset Metadata](#) [Dataset Structure Preview](#) [Dataset Quality Dashboard](#) [Dataset Uses](#)

Info
This page provides an overview of the dataset and associated study.

Overview of the study

The Artificial Intelligence Ready and Equitable Atlas for Diabetes Insights (AI-READI) project seeks to create a flagship ethically-sourced dataset to enable future generations of artificial intelligence/machine learning (AI/ML) research to provide critical insights into type 2 diabetes mellitus (T2DM), including salutogenic pathways to return to health. The ability to understand and affect the course of complex, multi-organ diseases such as T2DM has been limited by a lack of well-designed, high quality, large, and inclusive multimodal datasets. The AI-READI team of investigators will aim to collect a cross-sectional dataset of 4,000 people and longitudinal data from 10% of the study cohort across the US. The study cohort will be balanced for self-reported race/ethnicity, gender, and diabetes disease stage. Data collection will be specifically designed to permit downstream pseudo-time manifold analysis, an approach used to predict disease trajectories by collecting and learning from complex, multimodal data from participants with differing disease severity (normal to insulin-dependent T2DM). The long-term objective for this project is to develop a foundational dataset in T2DM, agnostic to existing classification criteria or biases, which can be used to reconstruct a temporal atlas of T2DM development and reversal towards health (i.e., salutogenesis). Data will be optimized for downstream AI/ML research and made publicly available. This project will also create a roadmap for ethical and equitable research that focuses on the diversity of the research participants and the workforce involved at all stages of the research process (study design and data collection, analysis, synthesis, and dissemination).

Usage statistics

6809 Views 0 Cited by 258 Access approved

All versions Current version

2.01 TB 165,051 Files

License

Health Data License

Keywords

Diabetes mellitus Machine Learning Artificial Intelligence Electrocardiography Continuous Glucose Monitoring Retinal imaging Eye exam

DOI

Accessible metadata

Indexed metadata

Clear access process

<https://doi.org/10.60775/fairhub.2>



Closing Remarks

Closing Remarks

Making data FAIR is not always easy...

> 1,000 standards

> 1,600 databases

> 100 policies

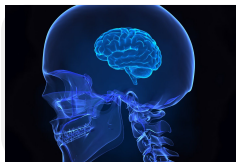


("Bio" related results on FAIRsharing.org)

Closing Remarks

... but you can get started today with simple steps!

Kids



Share/register dataset on a suitable repository or registry

[NIH list of data repositories](#)

Adults



Provide as much metadata as possible

For example: authors, funding source, keywords, devices used, etc.

Legends



Structure (meta)data following standards

[fairsharing.org](#)

Closing Remarks

FAIR is good, AI-ready is better!

New Results

Follow this preprint

AI-readiness for Biomedical Data: Bridge2AI Recommendations

Timothy Clark,

Harry Caufield,

Jillian A. Parker,

Sadnan Al Manir,

Edilberto Amorim,

James Eddy,

Nayoon Gim,

Brian Gow,

Wesley Goar,

Melissa Haendel,

Jan N. Hansen,

Nomi Harris,

Henning Hermjakob,

Marcin Joachimiak,

Gianna Jordan,

In-Hee Lee,

Shannon K. McWeeney,

Camille Nebeker,

Milen Nikolov,

Jamie Shaffer,

Nathan Sheffield,

Gloria Sheynkman,

James Stevenson,

Jake Y. Chen,

Chris Mungall,

Alex Wagner,

Sek Won Kong,

Satrajit S. Ghosh,

Bhaves Patel,

Andrew Williams,

Monica C. Munoz-Torres

the Bridge to Artificial Intelligence (Bridge2AI) Consortium members

doi: <https://doi.org/10.1101/2024.10.23.619844>

This article is a preprint and has not been certified by peer review [what does this mean?].

Abstract

Full Text

Info/History

Metrics

Preview PDF

Abstract

Biomedical research and clinical practice are in the midst of a transition toward significantly increased use of artificial intelligence (AI) and machine learning (ML) methods. These advances promise to enable qualitatively deeper insight into complex challenges formerly beyond the reach of analytic methods and human intuition while placing increased demands on ethical and explainable artificial intelligence (XAI), given the opaque nature of many deep learning methods.

<https://doi.org/10.1101/2024.10.23.619844>

Thank You!



bpatel@calmi2.org



fairdataihub.org

*Find these slides and
all resources here*



bit.ly/FAIR-AI-ready